# 8 Finance

## 8.6 Review of Finance

Formulas so far:

For these formulas, we have the following: $P_0$ is the principal. $r$ is the interest rate. $t$ is the number of time intervals (specified by problem) which have passed. $A$ is the amount paid back. $I$ is the interest.

| | |
|---|---|
| Simple Interest | $A = P_0(1+r)$ |
| Simple Interest over Time | $A = P_0(1+rt)$ |

For these formulas, we have the following: $P_0$ is the principal, or amount of a loan. $r$ is the annual interest rate. $t$ is the number of years which have passed. $A$ is the total amount. $k$ is the number of times interest is compounded per year. $d$ is the amount we deposit (or withdraw) from the account regularly, or the payment amount on a loan.

| | |
|---|---|
| Compound Interest | $A = P_0\left(1+\frac{r}{k}\right)^{kt}$ |
| Savings Annuity | $A = d\frac{\left(\left(1+\frac{r}{k}\right)^{kt}-1\right)}{\frac{r}{k}}$ |
| Payout Annuity/Loans | $P_0 = d\frac{\left(1-\left(1+\frac{r}{k}\right)^{-kt}\right)}{\frac{r}{k}}$ |

**Example 8.6.1** Suppose you obtain a \$3,000 T-note with a 3% annual rate, paid quarterly, with maturity in 5 years. How much interest will you earn?

**Example 8.6.2** You deposit \$300 in an account earning 5% interest compounded annually. How much will you have in the account in 10 years?

**Example 8.6.3** You want to be able to withdraw \$30,000 each year for 25 years. Your account earns 8% interest.

1. How much do you need in your account at the beginning
2. How much total money will you pull out of the account?
3. How much of that money is interest?

**Example 8.6.4** You want to buy a $25,000 car. The company is offering a 2% interest rate for 48 months (4 years). What will your monthly payments be?

**Example 8.6.5** A friend lends you $200 for a week, which you agree to repay with 5% one-time interest. How much will you have to repay?

**Example 8.6.6** You wish to have \$3000 in 2 years to buy a fancy new stereo system. How much should you deposit each quarter into an account paying 8% compounded quarterly?

**Example 8.6.7** You can afford a \$700 per month mortgage payment. You've found a 30 year loan at 5% interest.

1. How big of a loan can you afford?
2. How much total money will you pay the loan company?
3. How much of that money is interest?

**Example 8.6.8** You deposit \$1000 each year into an account earning 8% compounded annually.

1. How much will you have in the account in 10 years?
2. How much total money will you put into the account?
3. How much total interest will you earn?